BiteFixes Enterprise Architecture Manual

Volume 18

Disaster Recovery & Business Continuity Architecture (DRBC)

Version: 1.0

Classification: Enterprise Architecture

Status: Draft

Table of Contents

18.1 Purpose

Overview

The Disaster Recovery & Business Continuity Architecture defines how BiteFixes protects critical business operations and restores services after major disruptions.

The architecture ensures that the platform can recover from:

- Infrastructure failures.
- Cybersecurity incidents.
- Data corruption.
- Human errors.
- Cloud provider failures.
- Application failures.
- External dependency outages.

Strategic Objectives

The DRBC architecture aims to:

- Minimize service interruption.
- Protect customer data.
- Preserve business operations.
- Restore critical services quickly.
- Maintain customer trust.

- Support regulatory and enterprise requirements.

18.2 Disaster Recovery Vision

Overview

Disaster Recovery is not only about restoring servers.

For BiteFixes, recovery means restoring the complete business capability:

```
Infrastructure

        +

Applications

        +

Database

        +

AI Services

        +

Integrations

        +

Operational Processes

        =

Complete Platform Recovery
```

Recovery Philosophy

The platform follows the principle:

"Assume failure, prepare recovery."

Every critical component must have:

- Recovery procedure.
- Backup strategy.
- Responsible owner.
- Validation process.

18.3 Business Continuity Principles

Principle 1 — Critical Services First

Not all services have equal priority.

Recovery order follows business impact.

Example:

Priority 1:

```
Authentication

Customer Database

Ticket System

Core API
```

Priority 2:

```
AI Enhancement

Notifications
```

`Reports`

Priority 3:

`Analytics`

`Historical Processing`

Principle 2 — Data Protection

Customer information is considered a critical asset.

Protected data includes:

- Customer profiles.
- Tickets.
- Conversations.
- Devices.
- Repair history.
- AI memory.
- Company information.

Principle 3 — Automation

Recovery procedures should minimize manual intervention.

Automation includes:

- Backup verification.
- Service restoration.
- Infrastructure provisioning.
- Health validation.

Principle 4 — Continuous Improvement

Every recovery event generates learning.

After incidents:

- Analyze causes.
- Improve procedures.
- Update documentation.
- Improve architecture.

18.4 Disaster Recovery Architecture Overview

The recovery architecture uses multiple protection layers.

```
                Production Environment

                         │
                         ▼
                  Backup Systems

                         │
                         ▼
                 Recovery Environment

                         │
                         ▼
             Business Continuity Operations
```

Recovery Layers

Application Layer

Protection:

- Source control.
- Deployment automation.
- Configuration backup.

Data Layer

Protection:

- Database backups.
- Replication.
- Point-in-time recovery.

Infrastructure Layer

Protection:

- Infrastructure definitions.
- Environment documentation.
- Automated provisioning.

Integration Layer

Protection:

- API credentials management.
- Configuration backup.
- Alternative communication paths.

18.5 Disaster Classification

Disasters are classified according to impact.

Level 1 — Minor Incident

Limited impact.

Examples:

- Single service failure.
- Temporary API issue.
- Worker failure.

Recovery:

Minutes.

Level 2 — Major Incident

Multiple services affected.

Examples:

- Database degradation.
- Infrastructure failure.
- Integration outage.

Recovery:

Hours.

Level 3 — Critical Disaster

Business operations significantly affected.

Examples:

- Complete platform outage.
- Data corruption.
- Security incident.

Recovery:

Emergency response.

Level 4 — Catastrophic Event

Large-scale disruption.

Examples:

- Regional cloud failure.
- Major cyber event.
- Complete environment loss.

Recovery:

Full disaster recovery procedure.

18.6 Recovery Objectives (RTO/RPO)

Overview

Recovery objectives define how quickly services must return and how much data loss is acceptable.

Recovery Time Objective (RTO)

RTO defines maximum acceptable downtime.

Example:

```
Critical API

Target:
< 1 hour recovery


Non-critical Reports

Target:
< 24 hours recovery
```

Recovery Point Objective (RPO)

RPO defines maximum acceptable data loss.

Example:

```
Customer Data

RPO:
Minutes


Historical Analytics

RPO:
Hours
```

Service Recovery Classification

Service	Priority	Target

Authentication	Critical	Very Low RTO
Customer Database	Critical	Very Low RPO
Ticket System	Critical	Low RTO
Bitey AI	High	Medium RTO
Notifications	Medium	Higher RTO
Analytics	Low	Extended RTO

Chapter Summary

The first section of the **Disaster Recovery & Business Continuity Architecture** establishes the foundation for protecting BiteFixes against catastrophic events. It defines recovery philosophy, business continuity principles, disaster classification, and recovery objectives that guide all future protection strategies.

18.7 Backup Architecture

Overview

Backup Architecture defines how BiteFixes protects critical information against accidental deletion, corruption, security incidents, and infrastructure failures.

Backups are considered a fundamental security and continuity mechanism.

The backup strategy follows the principle:

"Data that cannot be restored is not protected."

Backup Objectives

The backup architecture provides:

- Data preservation.
- Historical recovery.
- Disaster restoration.
- Compliance support.

- Operational resilience.

Backup Categories

Database Backups

Protect transactional information.

Includes:

- Customers.
- Companies.
- Tickets.
- Conversations.
- Services.
- Devices.
- AI memory.

Application Backups

Protect software assets.

Includes:

- Source code.
- Configuration.
- Environment definitions.
- Deployment scripts.

Infrastructure Backups

Protect operational environments.

Includes:

- Server configuration.
- Network configuration.

- Security policies.
- Infrastructure templates.

Integration Backups

Protect external connectivity.

Includes:

- API configurations.
- Webhook definitions.
- Integration settings.
- Credentials metadata.

Backup Strategy

The platform follows a layered backup model.

```
Primary Environment

        |

        ▼

Automatic Backups

        |

        ▼

Backup Storage

        |

        ▼

Recovery Environment
```

Backup Frequency

Backup frequency depends on business importance.

Example:

Data Type	Frequency
Customer Data	Continuous / Frequent
Tickets	Frequent
Conversations	Frequent
Configuration	Daily
Analytics	Periodic

Backup Retention

Retention policies balance:

- Recovery requirements.
- Storage cost.
- Compliance needs.

Retention categories:

```
Short Term

↓

Medium Term

↓

Long Term Archive
```

Backup Validation

A backup is only considered valid after successful verification.

Validation includes:

- File integrity.
- Database consistency.
- Restore testing.
- Accessibility verification.

18.8 Data Recovery Strategy

Overview

Data Recovery defines how BiteFixes restores information after data loss or corruption.

Recovery Sources

Recovery may use:

- Latest backup.
- Point-in-time recovery.
- Database replicas.
- Archived data.
- Transaction logs.

Recovery Flow

```
Disaster Detected

        |

        ▼

Identify Recovery Point

        |
```

```
        ▼
Restore Data
        │
        ▼
Validate Integrity
        │
        ▼
Reconnect Services
        │
        ▼
Return To Operation
```

Data Recovery Priorities

Priority 1

Operational data:

- Customers.
- Authentication.
- Tickets.
- Conversations.

Priority 2

Business intelligence:

- Reports.
- Historical analytics.
- Metrics.

Priority 3

Non-critical historical information.

Data Integrity Validation

After recovery:

The platform verifies:

- Record consistency.
- Relationships.
- Permissions.
- Tenant isolation.
- Application compatibility.

18.9 Application Recovery Strategy

Overview

Application Recovery restores BiteFixes software services after failures.

Recovery Components

The recovery process includes:

`Source Code`

```
+

Dependencies

+

Configuration

+

Environment Variables

+

Deployment Pipeline

=

Application Recovery
```

Application Recovery Process

```
Environment Failure

        │

        ▼

Provision New Environment

        │

        ▼

Deploy Application

        │

        ▼
```

```
Restore Configuration
        |
        ▼
Execute Tests
        |
        ▼
Release Service
```

Deployment Recovery

The platform uses repeatable deployment processes.

Benefits:

- Faster restoration.
- Reduced human error.
- Consistent environments.

Configuration Recovery

Protected configuration includes:

- Database connections.
- API settings.
- Security parameters.
- Feature configuration.

Sensitive values are restored securely.

18.10 Infrastructure Recovery

Overview

Infrastructure Recovery restores the computing environment required for BiteFixes operation.

Infrastructure Components

Recovery covers:

- Compute resources.
- Networking.
- Storage.
- Security controls.
- Monitoring systems.

Infrastructure as Code Strategy

Future environments are defined through automated infrastructure descriptions.

Benefits:

- Repeatability.
- Version control.
- Faster disaster recovery.
- Reduced configuration drift.

Infrastructure Recovery Flow

```
Infrastructure Failure

          |
```

```
        ▼
Provision Resources
        │
        ▼
Apply Configuration
        │
        ▼
Deploy Services
        │
        ▼
Validate Health
        │
        ▼
Production Recovery
```

Environment Types

The platform maintains different environments:

Production

Customer-facing operations.

Staging

Testing and validation.

Development

Engineering activities.

18.11 Database Disaster Recovery

Overview

The database represents one of the highest priority recovery components.

Database recovery must preserve:

- Accuracy.
- Consistency.
- Tenant isolation.
- Transaction integrity.

Database Protection Layers

```
Primary Database

        │

        ▼

Replication

        │

        ▼
```

```
Backups
          │
          ▼
Archive Storage
```

Recovery Scenarios

Logical Failure

Examples:

- Accidental deletion.
- Incorrect update.

Recovery:

Point-in-time restoration.

Hardware Failure

Recovery:

Replica promotion or backup restoration.

Complete Environment Loss

Recovery:

Restore into new infrastructure.

Database Validation

After restoration:

Verify:

- Tables.
- Relationships.
- Indexes.
- Permissions.
- Application connectivity.

Chapter Summary

This section defines how BiteFixes protects and restores its most valuable assets: data, applications, and infrastructure. The backup and recovery strategy combines automated backups, validation procedures, repeatable deployments, and database protection mechanisms to minimize downtime and prevent irreversible data loss.

BiteFixes Enterprise Architecture Manual

Volume 18

Disaster Recovery & Business Continuity Architecture (DRBC)

18.12 AI System Recovery

Overview

Bitey represents a critical intelligence component of the BiteFixes platform. Recovery procedures must restore not only AI availability but also preserve AI knowledge, memory, configuration, and operational behavior.

The AI recovery strategy protects:

- AI Engine services.
- Knowledge Base.
- Intent models.
- Conversation memory.
- AI configuration.
- Prompt versions.
- Integration connectors.

AI Recovery Architecture

```
AI Runtime

     │

     ▼

AI Configuration

     │

     ▼

Knowledge Base

     │

     ▼
```

```
Conversation Memory
     │
     ▼
External AI Providers
```

AI Recovery Priorities

Priority 1 — Core AI Availability

Restore:

- AI processing services.
- Intent detection.
- Knowledge retrieval.
- Conversation handling.

Priority 2 — AI Knowledge

Restore:

- Technical articles.
- Service knowledge.
- Troubleshooting procedures.
- Business rules.

Priority 3 — AI Optimization Data

Restore:

- Historical conversations.
- Performance metrics.
- Learning information.

AI Failure Scenarios

AI Service Failure

Example:

Bitey processing unavailable.

Recovery:

- Restart AI workers.
- Redirect workload.
- Activate fallback processing.

Knowledge Base Failure

Recovery:

- Restore knowledge repository.
- Validate indexing.
- Rebuild search capability.

Memory Failure

Recovery:

- Restore conversation memory.
- Validate customer context.
- Re-enable personalization.

AI Graceful Degradation

During AI recovery:

```
Customer Request
```

```
        |
        ▼
Basic Support Flow
        |
        ▼
Ticket Creation
        |
        ▼
AI Processing Restored Later
```

The customer experience remains functional.

18.13 Integration Recovery

Overview

BiteFixes depends on external ecosystems such as:

- WhatsApp.
- WordPress.
- Payment platforms.
- Email services.
- Notification providers.

Integration Recovery ensures business continuity when external systems fail.

Integration Recovery Principles

Dependency Isolation

External failures must not stop the core platform.

Example:

```
WhatsApp Failure

        |

        ▼

WhatsApp Channel Disabled

        |

        ▼

Website Chat Continues

Ticket System Continues
```

Integration Recovery Process

```
Integration Failure

        |

        ▼

Detect Failure

        |

        ▼
```

```
Activate Fallback
        │
        ▼
Queue Pending Operations
        │
        ▼
Restore Connection
        │
        ▼
Synchronize Data
```

Webhook Recovery

Webhook protection includes:

- Retry mechanisms.
- Event storage.
- Duplicate prevention.
- Delivery tracking.

External API Recovery

For third-party APIs:

The platform manages:

- Timeouts.
- Retries.
- Circuit breakers.

- Alternative workflows.

18.14 Business Continuity Operations

Overview

Business Continuity ensures that BiteFixes continues delivering value even during major disruptions.

Recovery is not only technical; it includes operational procedures.

Business Continuity Model

```
Technology Recovery

        +

Operational Recovery

        +

Customer Communication

        +

Management Coordination

        =

Business Continuity
```

Critical Business Functions

Customer Support

Must maintain:

- Ticket creation.
- Customer communication.
- Service tracking.

Technical Operations

Must maintain:

- Diagnostics.
- Repair workflow.
- Technician coordination.

Administration

Must maintain:

- User management.
- Company management.
- Billing operations.

Communication Strategy

During incidents:

Stakeholders receive:

- Incident status.
- Expected impact.
- Recovery progress.
- Resolution confirmation.

Business Continuity Roles

Platform Administrator

Responsible for:

- Recovery execution.
- System validation.
- Technical coordination.

Operations Manager

Responsible for:

- Business impact.
- Customer communication.
- Priority decisions.

Technical Team

Responsible for:

- Diagnosis.
- Restoration.
- Verification.

18.15 Disaster Recovery Testing

Overview

A recovery plan that is never tested cannot be considered reliable.

BiteFixes performs regular recovery validation.

Testing Types

Backup Restoration Test

Validates:

- Backup integrity.
- Restore procedures.
- Data availability.

Application Recovery Test

Validates:

- Deployment process.
- Configuration restoration.
- Service startup.

Full Disaster Simulation

Simulates:

- Environment loss.
- Major failure.
- Complete recovery process.

Testing Frequency

Testing frequency depends on importance.

Example:

```
Critical Systems
```

```
Frequent Testing

Secondary Systems

Periodic Testing

Archive Systems

Scheduled Validation
```

Recovery Metrics

Measured indicators:

RTO Achievement

Was recovery completed within target time?

RPO Achievement

Was data loss within acceptable limits?

Recovery Success Rate

Did the process complete successfully?

Improvement Actions

Each test produces:

- Findings.

- Corrections.
- Updated procedures.

18.16 Future Evolution

Overview

The Disaster Recovery architecture evolves toward automated resilience.

Future capabilities include:

Automated Disaster Detection

AI systems will detect:

- Infrastructure failures.
- Abnormal behavior.
- Security events.

Autonomous Recovery

Future platform capabilities:

```
Failure Detected

        |

        ▼

AI Analysis

        |

        ▼
```

```
Recovery Decision

        │

        ▼

Automatic Restoration

        │

        ▼

Validation
```

Multi-Region Disaster Recovery

Future global deployments may support:

```
Primary Region

        │

        ▼

Secondary Region

        │

        ▼

Global Recovery Capability
```

Continuous Resilience Engineering

The platform evolves through:

- Regular testing.
- Architecture improvement.

- Automation.
- Failure simulation.

18.17 Volume Summary

The **Disaster Recovery & Business Continuity Architecture (DRBC)** establishes how BiteFixes protects operations, data, AI capabilities, and customer services during critical events.

The architecture defines:

- Backup strategies.
- Data recovery.
- Application restoration.
- Infrastructure recovery.
- AI recovery.
- Integration resilience.
- Business continuity procedures.
- Disaster testing.
- Future autonomous recovery.

The central principle is:

A reliable SaaS platform is not defined by the absence of failures, but by the ability to recover quickly and safely.

Through layered protection, automation, redundancy, and continuous improvement, BiteFixes achieves the resilience required for enterprise-scale operations.